Sunny Zhang

647-328-2656 | ssunny.zhang@mail.utoronto.ca | linkedin.com/in/sunnyzhang8 | sunnyzhang.dev

Education

University of Toronto, St. George — Toronto, ON
Honours BSc, Double Major in Computer Science and Mathematics — *Sep 2025 – Apr 2029*

Experience

AI Research Intern — Jun 2026 – Sep 2026
Algoverse AI Research Program — *Remote*

- Led a 3-person team, came up with the architecture and implemented a 4-agent Llama-3.3-70B trading simulator in Python; ran 120 episodes that produced 2,400 labeled decisions.
- Built a seeded experiment harness for 5 pressure levels and 2 authority conditions; logged the messages and capital decisions separately to compare LLM language with its actions.
- The experiment found that managers passed on capital threats in 0.3% of messages they sent compared to 32.0% they received, while the trader capital gap grew from 0.55 to 0.85.

Projects

LaunchPilot — **Hack Canada 2x Track Winner** | *Next.js, TypeScript, FastAPI, Postgres, Docker*

- Built an AI launch assistant in 36 hours: 4 agents research a market, write positioning, generate assets, queue outreach emails behind a human approval gate, and build a 7-day launch plan.
- Owned the agent orchestration and the inter-agent communication pipeline, threading shared project context through persistent stage threads so each agent builds on the context of the previous agents instead of restarting cold.
- Developed the auth system end to end: Auth0 JWT validation, scope-gated API routes, and the frontend session flow, plus a dev mode keeping the 49-endpoint API runnable without an Auth0 tenant.

Tiny Society | *Next.js, TypeScript, FastAPI, Supabase, Docker*

- Built a full-stack multi-agent simulator where up to 7 characters with ranked memory recall, plans, and periodic reflection interact in a user-authored world that is streamed to a UI where players can step through day by day.
- Designed a deterministic relationship model that maps agent output to a calibrated effect so bonds build gradually, romances and alliances form only when mutual, and out-of-character actions are blocked.
- Developed a "World Lens" that holds a world's premise (constraints, gathering places, narrative tone, and central conflict) and reuses it everywhere, eliminating repeated interpretation while keeping stories true to the premise.

Feedback or Annealing? [Preprint] | *Python, Numba, pytest, mypy*

- Built a reproducible study of adaptive state aggregation for dynamic programming on MDPs: 3 width rules benchmarked on a 9,261-state inventory MDP across 20 shared seeds and 5 compute budgets.
- Wrote the solver as Numba JIT kernels using flat CSR-style transition arrays: one Bellman backup specialized at compile time for per-state or per-group lookups, plus a counting-sort rebin into preallocated buffers.
- Showed plain value iteration beat every aggregation method: it converged in 4.2M backups while each aggregation run spent 9.2M+, and the residual feedback signal improved error by at most 0.2%.

Publications

1. **Sunny Zhang**, Francesco Febbo, Harshit Saini. *Language–Action Decoupling: Speech as a Weak Proxy for Action in LLM Hierarchies.* **Accepted to four NeurIPS 2026 workshops:** IAB, AIWILD, VerifyAgents, SocialAgent. sunnyzhang.dev/language-action-decoupling

Technical Skills

Languages: Python, TypeScript, SQL, Java, C, C++
Web: FastAPI, REST APIs, Flask, Next.js, React, PostgreSQL, SQLAlchemy, Alembic
Testing & Infrastructure: pytest, GitHub Actions, Docker, Git, Microsoft Azure, Fly.io, Vercel, Supabase
ML & Data: PyTorch, NumPy, Pandas, Numba